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CHAPTER – 6

MEASURES OF DISPERSION

6.1. Introduction

Dispersion is the second important characteristic of a frequency distribution. Two distributions may have the same mean, median and mode, but the variation of the individual observations of the two distributions may differ. For example, the daily wages in Taka of seven workers of two factories are as follows:

Wages of factory : I	142	143	150	150	153	155	157
Wages of factory : II	122	140	150	150	154	159	175

Here it is seen that the mean, median and modal wages of both the factories are same. It is Tk. 150. That is both the distributions have the same measures of location. But the individual observations are not same. The range of the wage structure of the first factory is Tk. 15 = 157 - 142 only whereas the range of the second factory is Tk. 53 = 175 - 122. That is the wage structure of the factory I is more compact than the wage structure of the factory II. The variability or dispersion of the individual values of factory 1 is less than the factory 2. Suppose the depths of a river at 5 different points are 4, 3, 2, 3, 8 feet. The average depth of a river at different points is 4ft; it would not give you the guarantee to cross the river safely. To cross the river safely, you have to know the maximum depth of the river. This means measures of location have failed to measure the variability of a set of data. The need for a measure of dispersion in addition to a measure of location is necessary. Small dispersion indicates high uniformity of the observations in the distribution.

Dispersion tells us how compactly the individual values are distributed around the central values. Dispersion of a single variable might not bear that much meaning, while comparison of dispersion of two sets of variables is more useful for taking decision.

According to Books and Dicks "Dispersion or spread is the degree of the scatter or variation of the variables about a central value.

Definition.6.1. Dispersion measures the variability of a set of observations among themselves or about some central values.

6.2. Purposes of Dispersion

Measure of dispersion is needed for four basic purposes :

i) To determine the reliability of an average.
ii) To serve as a basis for the control of the variability.
iii) To compare two or more series with regard to their variability.
iv) To facilitate the computation of other statistical measures.

6.3. Properties of a Good Measures of Dispersion

The properties of a good measure of dispersion are the same as those of a good measure of central tendency. They are as follows:

i) It should be simple to understand,
ii) It should be easy to compute,
iii) It should be rigidly defined,
iv) It should be based on all the observations,
v) It should have sampling stability,
vi) It should be suitable for further algebraic treatment, and
vii) It should not be affected by extreme observations.

6.4. Measures of Dispersion

The numerical values by which we measure the dispersion or variability of a data set or a frequency distribution are called measures of dispersion.

There are two kinds of measures of dispersion:

i) Absolute measures of dispersion and
ii) Relative measures of dispersion.

Important absolute measures of dispersion are:

i) Range,
ii) Quartile deviation,
iii) Mean deviation,
iv) Variance and Standard deviation.

The relative measures of dispersion are:

i) Coefficient of range,
ii) Coefficient of quartile deviation,
iii) Coefficient of mean deviation,
iv) Coefficient of variation.

It is seen that for each absolute measure of dispersion, there is a relative measure of dispersion. Absolute measures are expressed in the same unit in which the original variables are measured. For example, any absolute measure of weight may be measured as kilogram or pound, height as meter or inch, price as taka or dollar etc. On the other hand, relative measures are pure number and independent of the unit of measurement and express in percentage.

6.9. Coefficient of Variation

Standard deviation is an absolute measure of dispersion. The corresponding relative measure is known as the coefficient of variation. This measure developed by Karl Pearson is the most commonly used measure of relative dispersion.

Definition. If μ is the mean and σ is the standard deviation of a population data set, then coefficient of variation denoted by C.V. is defined by

$$C.V. = \frac{\sigma}{\mu} \times 100.$$

If $\bar{x}$ is the mean and s is the standard deviation of a sample data set, then coefficient of variation is defined by

$$C.V. = \frac{s}{\bar{x}} \times 100.$$

It is a pure number and expressed as a percentage. It is useful in comparing the variability of two or more sets of data, especially if they are expressed in different units of measurement. Since it is a ratio, the units of measurement have no significance. Moreover, coefficient of variation depends on the best measure of central tendency and the best measure of dispersion. In these reasons, coefficient of variation is better than standard deviation as a measure of dispersion. Overall, coefficient of variation is the best measure of dispersion among all the absolute and relative measures.

Comment. Coefficient of variation is the best measure of dispersion.

Example 6.9.1. The grade point averages obtained by randomly selected 6 students in their H.S.C. examination are as follows: 4.9, 4.1, 4.4, 3.3, 4.6 and 4.8. Find mean, standard deviation and coefficient of variation.

Solution. We know from the problem 6.8.3 that the standard deviation of the data set is 0.59 and $\Sigma x = 26.1$. Then $\bar{x} = \frac{26.1}{6} = 4.35$.

$$\text{Hence, C.V.} = \frac{s}{\bar{x}} \times 100 = \frac{0.59}{4.35} \times 100 = 13.56.$$

Among two or more data sets, the set for which coefficient of variation is greater is said to be more variable or conversely less consistent, less uniform, less stable or less homogenous. On the other hand, the series for which coefficient of variation is less is said to be less variability or more consistent, more uniform, more stable or more homogenous.

Example 6.9.2. Lives of two models of refrigerators in a recent survey were found as follows:

Life (no. of years)	Model A	Model B
0-2	5	2
2-4	16	7
4-6	13	12
6-8	7	19
8-10	5	9
10-12	4	1

i) What is the average life of each model of these refrigerators?
ii) Which of the two models shows more uniformity?
iii) A person wants to buy a new refrigerator, which one will he prefer?

Solution. For finding the average lifetimes, we have to compute arithmetic mean and for determining the model which has greater uniformity, then compute and compare the coefficient of variations.

Class interval	Mid-points x	Model A			Model B		
		f	fx	fx^2	f	fx	fx^2
0-2	1	5	5	5	2	2	2
2-4	3	16	48	144	7	21	63
4-6	5	13	65	325	12	60	300
6-8	7	7	49	343	19	133	931
8-10	9	5	35	405	9	81	729
10-12	11	4	44	484	1	11	121
Total		50	256	1706	50	308	2146

Computations of mean, variances and co-efficient of variations of lifetimes for two models are shown below:

	Model A	Model B
Arithmetic Mean,	$\bar{x}_A = \frac{256}{50} = 5.12$ years	$\bar{x}_B = \frac{308}{50} = 6.16$ years
	$s_A^2 = \frac{1}{49}\left[1706 - \frac{(256)^2}{50}\right]$	$s_B^2 = \frac{1}{49}\left[2146 - \frac{(308)^2}{50}\right]$
	$= \frac{1}{49}[1706 - 1310.72]$	$= \frac{1}{49}[2146 - 1897.28]$
	$= \frac{395.28}{49} = 8.07$	$= \frac{248.72}{49} = 5.08$
	$s_A = \sqrt{8.07} = 2.84$ years	$s_B = \sqrt{5.08} = 2.25$ years
	C.V. (A) $= \frac{2.84}{5.12} \times 100 = 55.47\%$	C.V.(B) $= \frac{2.25}{6.16} \times 100 = 36.53\%$.

i) Average lifetimes of refrigerators of Model A is 5.12 years, while of Model B is 6.16 years.

ii) Since coefficient of variation is less for Model B, hence refrigerators of Model B show greater uniformity as per the lifetime of the refrigerators.

iii) Due to the greater uniformity in lifetime, the person will prefer Model B.

Example 6.9.3. The following are some of the particulars of the distribution of weights of boys and girls in a class.

	Boys	Girls
Number :	65	35
Mean weight :	60kgs	45kgs
Standard deviation :	4kgs	2kgs

The weights of which distribution is more homogenous?

Solution. We can consider the data from two populations. To compare the variability of the weights of two distributions, we will have to find the coefficient of variations of the two distributions.

Boys

$$C.V. = \frac{\sigma}{\mu} \times 100 = \frac{4}{60} \times 100 = 6.67\%.$$

Girls

$$C.V. = \frac{\sigma}{\mu} \times 100 = \frac{2}{45} \times 100 = 4.44\%$$

Comment. Since the coefficient of variation is greater for boys, hence the weights of the boys is more variability and less homogenous.

Example 6.9.4. The following are some of the particulars of the distributions of weights and heights of 100 boys in a class.

	Weights	Heights
Mean weight :	60kgs	65 inches
Standard deviation :	7kgs	4 inches

Which distribution shows greater variability?

Solution. For determining variability between the two distributions, we have to compute coefficient of variations of weights and heights of the boys.

$$C.V.\ (Weight) = \frac{\sigma}{\mu} \times 100 = \frac{7}{60} \times 100 = 11.67\%$$

$$C.V.\ (Heights) = \frac{\sigma}{\mu} \times 100 = \frac{4}{65} \times 100 = 6.15\%$$

Comment. Since coefficient of variation is higher for weights, hence the distribution of weights shows greater variability than heights.

Example 6.9.5. In two factories A and B engaged in the same industry, the average weekly wages and standard deviations are as follows:

	Factory A	Factory B
Average weekly wages (in Taka) :	860	900
Standard deviation of wages (in Taka) :	50	80

i) Which factory shows greater variability in the distribution of wages?

ii) A person got job in both the factories, which factory he will prefer to join?

Solution. For comparing the variability of wages of the two factories, we have to compute coefficient of variations.

Factory A

$$C.V. = \frac{\sigma}{\mu} \times 100 = \frac{50}{860} \times 100 = 5.81\%$$

Factory B

$$C.V. = \frac{\sigma}{\mu} \times 100 = \frac{80}{900} \times 100 = 8.89\%.$$

Comment. The coefficient of variation is greater for factory B; hence factory B shows greater variability in the distribution of wages. The person will join factory A, since the coefficient of variation of the factory A is less than the factory B, although the average weekly wages of factory B is more than A.

Example 6.9.6. Two cricketers scored the following runs in randomly selected 10 one- day matches:

Player A	42	32	40	45	17	83	59	64	76	72
Player B	95	3	28	70	31	14	82	0	59	108

i) Who is the better run –getter?

ii) Who is the consistent player?

iii) A prize is given to the best player. Who will get the prize?

Solution. In order to find out who is better run-getter, we will compare the average runs scored and to find out who is more consistent, we will compare the coefficient of variations.

Calculation of Mean and Coefficient of variation

Cricketer A : x	x^2	Cricketer B : x	x^2
42	1764	95	9025
32	1024	3	9
40	1600	28	784
45	2025	70	4900
17	289	31	961
83	6889	14	196
59	3481	82	6724
64	4096	0	0
76	5776	59	3481
72	5184	108	11664
530	32127	490	37744

Cricketer A	Cricketer B
$\bar{x}_A = \frac{\Sigma x}{n} = \frac{530}{10} = 53$	$\bar{x}_B = \frac{\Sigma x}{n} = \frac{490}{10} = 49$

(i) Since the average score for the player A is higher than player B, hence A is a better run-getter. However,

$$s_A^2 = \frac{1}{9}\left(32127 - \frac{(530)^2}{10}\right) \qquad s_B^2 = \frac{1}{9}\left(37744 - \frac{(490)^2}{10}\right)$$

$$= \frac{1}{9}(32127 - 28090) = 448.56 \qquad = \frac{1}{9}(37744 - 24010) = 1526$$

Hence $s_A = 21.18$ Hence $s_B = 39.06$

$$C.V.(A) = \frac{21.18}{53} \times 100 = 39.96\% \qquad C.V.(B) = \frac{39.06}{49} \times 100 = 79.71\%.$$

(ii) The coefficient of variation for player A is less than player B; hence player A is more consistent.

(iii) The player A will get the prize.

6.9.1. Advantage of coefficient of variation over standard deviation. Standard deviation and the original variable have the same unit of measurements. So standard deviation cannot be used to compare the variability of two or more distributions measured in different units. But coefficient of variation is a pure number. That is coefficient of variation is independent of the unit of measurement of the original variables. Therefore, coefficient of variation can be successfully used to compare the variability of two or more distributions in different units of measurement. Hence coefficient of variation is sometimes better than standard deviation as measure of dispersion.

6.9.2. Practical Significance of the standard deviation (Chebyshev's Theorem). The two summary values most often used by statisticians are the mean and standard deviation. If the observations of a data set have a small standard deviation, we would expect most of the values to be grouped around the mean. However, a large value of the standard deviation indicates a greater variability, and therefore we would expect the observations to be more spread out the mean. The Russian mathematician P.L.Chebyshev (1821-1894) discovered that the fraction of the observations falling between any two values symmetric about the mean is related to the standard deviation. Chebyshev's theorem gives a conservative estimate of the fraction of measurements falling within k standard deviation of the mean, that means within $(\mu - k\sigma)$ to $(\mu + k\sigma)$, in usual notations.

Chebyshev's Theorem.

Given a number k greater than or equal to 1, at least the fraction $(1 - 1/k^2)$ of the observations of any set of data lies within k standard deviations of the mean of the observations.

Chebyshev's theorem can be applied to any set of data and can be used to describe either a sample or a population.

Now let us choose a few numerical values for k and compute $(1 - 1/k^2)$.

k	$(1-1/k^2)$
1	$1-1=0$
2	$1-\frac{1}{4}=\frac{3}{4}$
3	$1-\frac{1}{9}=\frac{8}{9}$

Thus, the theorem states that, in case of a Population:

1. For k =1, at least none of the observations lie in the interval $\mu-\sigma$ to $\mu+\sigma$.
2. For k=2, at least 3/4 or 75% of the observations lie in the interval $\mu-2\sigma$ to $\mu+2\sigma$.
3. For k=3, at least 8/9 or 89% of the observations lie in the interval $\mu-3\sigma$ to $\mu+3\sigma$.

And in case of a Sample

1. For k = 1, at least none of the observations lie in the interval $\bar{x}-s$ to $\bar{x}+s$.
2. For k = 2, at least 3/4 or 75% of the observations lie in the interval $\bar{x}-2s$ to $\bar{x}+2s$.
3. For k = 3, at least 8/9 or 89% of the observations lie in the interval $\bar{x}-3s$ to $\bar{x}+3s$.

The first statement does not provide any valuable information about the set of observations, but the other two values of k provide valuable information about the proportion of observations that fall in certain intervals. One can use any value of k for finding the desired proportion of observations, for example, the proportion of observations that fall within k = 2.5 standard deviations of the mean is at least $1 - 1/(2.5)^2 = 0.84$ or 84%.

Example 6.9.7. A population of 180 observations has a mean of 32 and a standard deviation of 4. According to Chebyshev's theorem

(a) How many of the observations fall in the interval from 24 to 40?